Department of Applied Geology, IIT(ISM) Dhanbad
Monsoon Semester 2023-24; 3rd Semester B. Tech
Lecture Plan: Geology for Engineering and Sciences (GLE201)

Course Type	Course Code	Name of Course	L	T	P	Credit
E/SO	GLE 201	Geology for Engineering and Sciences	3	0	0	9

Course Objective
The primary objective of the course is to introduce fundamental concepts, ideas, and materials in geology to students of science and engineering.
Learning Outcomes
Upon completion of the course, students will be able to: 1. Understand the basics of mineralogy and petrology. 2. Learn about the fundamentals of palaeontology and stratigraphy. 3. Understand physical and structural geology.

Unit No.	Topics to be Covered	Lecture Hours	Learning Outcome
1.	Mineralogy: General properties; Bowen's Reaction Series, Classification of minerals and properties of common rock forming minerals; Megascopic identification of some rock forming minerals	7	This unit will help the student in understanding the fundamentals and mineralogy.
2.	Petrology: Rock cycle, Rock types, Classification and description of some common rocks; Megascopic identification of igneous, sedimentary and metamorphic rocks	6	This unit will help the student in learning petrology of igneous, sedimentary and metamorphic rocks.
3.	Stratigraphy and Palaeontology: Principles of stratigraphy; Geologic Time Scale; Broad stratigraphic subdivisions and associated rock types of important coal belts and oil fields of India; Concepts of palaeontology; Fossils and their mode of preservation; Concept of index fossils	7	This unit will help the student in learning the fundamentals of stratigraphic classification and paleontology.
4.	Mineral and Energy Resources: Introduction and scope of economic geology (including coal and hydrocarbon resources); Ore and gangue minerals; Resource, reserve, and grade; Distribution and mode of occurrence of some mineral deposits, coal and petroleum deposits; Megascopic identification of some ore-forming minerals	5	This unit will help the student in understanding different mineral and energy resources and their distribution and mode of occurrence.
5.	Physical Geology: Evolution of the earth; Exogenous and endogenous processes shaping the Earth; Important geomorphological features	7	This unit will help the student in understanding exogenous and endogenous processes that shape the Earth.

6.	Structural Geology: Interpretation of topographic and geological maps; Attitude of planar and linear structures; Effects of topography on outcrops; Unconformities, folds, faults and joints - their nomenclature, classification and recognition; Some structural geological problems and their solutions	**7**	This unit will help the student to understand aspects of structural geology and learn to solve structural geological problems.
Total Classes		**39**	

Text Books:

1. Hefferan, K. and O'Brien, J., 2010. Earth Materials, Wiley-Blackwell, Sussex; 670 p.
2. Mukherjee, P.K., A textbook of geology. World press.

Reference Books:
1. Jain, S., 2014. Fundamentals of Physical Geology, Springer, New Delhi; 494 p.
2. Davis, G.H., Reynolds, S.J., 1996. Structural Geology of Rocks and Regions, John Wiley & Sons, Inc., New York; 776 p.

Evaluation Procedure (For Theory Course):
The breakup of distribution of marks for evaluation of the performance of students (out 100 marks) in the course, is:

Details	**Marks**
Quiz/Assignment	20
Mid Semester	32
End Semester Examination	48
Total Marks	**100**

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